Inverse Trigonometric Functions: Questions and Full Solutions

Question 01

English

Let $\alpha = -\dfrac{1+\sqrt{5}}{4}$. Considering only the principal values of the inverse trigonometric functions, the value of $\sin^{-1}\alpha + \cos^{-1}\left(2\alpha^2 - 1\right)$ is

(A) $-\dfrac{\pi}{5}$

(B) $\dfrac{\pi}{5}$

(C) $\dfrac{\pi}{10}$

(D) $\dfrac{13\pi}{10}$

Only one option is correct.

हिन्दी

माना कि $\alpha = -\dfrac{1+\sqrt{5}}{4}$ है। प्रतिलोम त्रिकोणमितीय फलनों के केवल मुख्य मानों को लेते हुए, $\sin^{-1}\alpha + \cos^{-1}\left(2\alpha^2 - 1\right)$ का मान है

(A) $-\dfrac{\pi}{5}$

(B) $\dfrac{\pi}{5}$

(C) $\dfrac{\pi}{10}$

(D) $\dfrac{13\pi}{10}$

केवल एक विकल्प सही है।

Solution

The number $\dfrac{1+\sqrt{5}}{4}$ is the standard value $\cos 36^\circ$ (equivalently, $\sin 18^\circ = \dfrac{\sqrt{5}-1}{4}$ and $\cos 36^\circ = 1 - 2\sin^2 18^\circ = \dfrac{1+\sqrt{5}}{4}$). Hence $\alpha = -\cos 36^\circ = -\sin 54^\circ$.

Since $\sin^{-1}$ is odd and $54^\circ = \dfrac{3\pi}{10}$ lies in $\left[-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right]$, $\sin^{-1}\alpha = -\sin^{-1}(\sin 54^\circ) = -\dfrac{3\pi}{10}$. (Taking $\sin^{-1}\alpha = -\dfrac{\pi}{5}$ confuses $\cos 36^\circ$ with $\sin 36^\circ$.)

Next, $\alpha^2 = \cos^2 36^\circ$, so $2\alpha^2 - 1 = \cos 72^\circ$. Directly: $\alpha^2 = \dfrac{6+2\sqrt{5}}{16}$, so $2\alpha^2 - 1 = \dfrac{6+2\sqrt{5}}{8} - 1 = \dfrac{\sqrt{5}-1}{4} = \sin 18^\circ = \cos 72^\circ$.

Because $72^\circ = \dfrac{2\pi}{5} \in [0, \pi]$, $\cos^{-1}\left(2\alpha^2 - 1\right) = \dfrac{2\pi}{5}$. The identity $\cos^{-1}(2x^2 - 1) = 2\cos^{-1} x$ holds only for $x \in [0, 1]$; here $\alpha < 0$ and $2\cos^{-1}\alpha = 2 \cdot \dfrac{4\pi}{5} = \dfrac{8\pi}{5}$ is outside the principal range, so it cannot be used.

Therefore $\sin^{-1}\alpha + \cos^{-1}\left(2\alpha^2 - 1\right) = -\dfrac{3\pi}{10} + \dfrac{2\pi}{5} = \dfrac{\pi}{10}$.

Correct Option: (C)

Question 02

English

For each positive integer n, let $S_n = \sum_{k=1}^{n} \tan^{-1}\left(\frac{4k}{k^4 - k^2 + 4}\right)$, where $\tan^{-1}$ takes values in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then $S_2 + S_3$ is equal to

(A) $\pi - \tan^{-1}\left(\frac{9}{17}\right)$

(B) $\pi - \tan^{-1}\left(\frac{17}{9}\right)$

(C) $\frac{3\pi}{4}$

(D) $\pi - \tan^{-1}\left(\frac{9}{19}\right)$

Only one option is correct.

हिन्दी

प्रत्येक धनात्मक पूर्णांक n के लिए, माना कि $S_n = \sum_{k=1}^{n} \tan^{-1}\left(\frac{4k}{k^4 - k^2 + 4}\right)$ है, जहाँ $\tan^{-1}$ के मान $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ में हैं। तब $S_2 + S_3$ बराबर है

(A) $\pi - \tan^{-1}\left(\frac{9}{17}\right)$

(B) $\pi - \tan^{-1}\left(\frac{17}{9}\right)$

(C) $\frac{3\pi}{4}$

(D) $\pi - \tan^{-1}\left(\frac{9}{19}\right)$

केवल एक विकल्प सही है।

Solution

Write $T_k = \frac{k(k+1)}{2}$ (so $T_0 = 0, T_1 = 1, T_2 = 3, T_3 = 6$). Then $T_k - T_{k-1} = \frac{k(k+1) - (k-1)k}{2} = k$ and $T_k T_{k-1} = \frac{k(k+1)}{2} \cdot \frac{k(k-1)}{2} = \frac{k^2(k^2-1)}{4}$.

Hence $1 + T_k T_{k-1} = \frac{k^4 - k^2 + 4}{4}$ and $\frac{4k}{k^4 - k^2 + 4} = \frac{k}{1 + \frac{k^2(k^2-1)}{4}} = \frac{T_k - T_{k-1}}{1 + T_k T_{k-1}}$.

For $a, b \geq 0$ we have $ab > -1$, so the subtraction formula $\tan^{-1} a - \tan^{-1} b = \tan^{-1} \frac{a-b}{1+ab}$ holds without correction. Therefore $\tan^{-1}\left(\frac{4k}{k^4 - k^2 + 4}\right) = \tan^{-1} T_k - \tan^{-1} T_{k-1}$.

Adding for $k = 1, 2, \ldots, n$, the sum telescopes: $S_n = \tan^{-1} T_n - \tan^{-1} T_0 = \tan^{-1} \frac{n(n+1)}{2}$. Check: $S_1 = \tan^{-1} 1 = \frac{\pi}{4}$, and the first term is $\tan^{-1} \frac{4}{4} = \frac{\pi}{4}$.

Thus $S_2 = \tan^{-1} 3$ and $S_3 = \tan^{-1} 6$.

For the addition, both numbers are positive and $3 \cdot 6 = 18 > 1$, so $\tan^{-1} 3 + \tan^{-1} 6 > \frac{\pi}{2}$ and the addition formula needs the correction π: $\tan^{-1} 3 + \tan^{-1} 6 = \pi + \tan^{-1} \frac{3+6}{1-18} = \pi + \tan^{-1}\left(-\frac{9}{17}\right) =$

$\pi - \tan^{-1}\dfrac{9}{17}$.

(Forgetting the correction gives the negative value $-\tan^{-1}\dfrac{9}{17}$, impossible for a sum of positive angles.)

Correct Option: (A)

Question 03

English

Let S denote the set of all real numbers x satisfying $2\sin^{-1}(\sin x) = \cos^{-1}\left(\dfrac{x}{20}\right)$, where the inverse trigonometric functions take their principal values. Then the number of elements in S is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि S उन सभी वास्तविक संख्याओं x का समुच्चय है जो $2\sin^{-1}(\sin x) = \cos^{-1}\left(\dfrac{x}{20}\right)$ को संतुष्ट करती हैं, जहाँ प्रतिलोम त्रिकोणमितीय फलन अपने मुख्य मान लेते हैं। तब S में अवयवों की संख्या _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Domain. $\cos^{-1}\dfrac{x}{20}$ is defined only for $-20 \le x \le 20$; $\sin^{-1}(\sin x)$ is defined for all x. So every solution lies in $[-20, 20]$.

Range. $\cos^{-1}\dfrac{x}{20} \in [0, \pi]$, so a solution needs $2\sin^{-1}(\sin x) \in [0, \pi]$, that is $f(x) = \sin^{-1}(\sin x) \in \left[0, \dfrac{\pi}{2}\right]$. Write the equation as $f(x) = g(x)$ with $g(x) = \dfrac{1}{2}\cos^{-1}\dfrac{x}{20}$.

Graph of f: a triangle wave of period 2π with $f(x) = x - 2k\pi$ on $\left[2k\pi - \dfrac{\pi}{2}, 2k\pi + \dfrac{\pi}{2}\right]$ and $f(x) = (2k+1)\pi - x$ on $\left[2k\pi + \dfrac{\pi}{2}, 2k\pi + \dfrac{3\pi}{2}\right]$. So $f \ge 0$ exactly on the humps $[2k\pi, (2k+1)\pi]$, rising from 0 to $\dfrac{\pi}{2}$ and back to 0.

Graph of g: strictly decreasing from $g(-20) = \dfrac{\pi}{2}$ to $g(20) = 0$, and $0 < g < \dfrac{\pi}{2}$ for $-20 < x < 20$. Its slope is $g'(x) = -\dfrac{1}{2\sqrt{400 - x^2}}$, with $|g'| < 1$ whenever $\sqrt{400 - x^2} > \dfrac{1}{2}$, that is for all $|x| < 19.99$.

Full humps in $[-20, 20]$: $2k\pi \ge -20$ and $(2k+1)\pi \le 20$ give $k = -3, -2, -1, 0, 1, 2$, i.e. $[-6\pi, -5\pi], [-4\pi, -3\pi], [-2$ (note $-6\pi \approx -18.85 > -20$ and $5\pi \approx 15.71$).

On such a hump, $h = f - g$ satisfies $h = -g < 0$ at both ends and $h = \dfrac{\pi}{2} - g > 0$ at the midpoint. On the rising half $h' = 1 - g' > 0$; on the falling half $h' = -1 - g' = -1 + |g'| < 0$. So h has exactly one zero on each half: 2 solutions per hump, 12 in all.

Partial hump $[6\pi, 20]$ ($6\pi \approx 18.85$, $7\pi > 20$): here $f(x) = x - 6\pi$ increases from 0 to $20 - 6\pi \approx 1.15 < \dfrac{\pi}{2}$, while g decreases from $g(6\pi) \approx 0.17$ to 0. Then h is increasing, negative at 6π and positive at 20: exactly 1 solution.

On $[-20, -6\pi)$, $x + 6\pi \in [-1.15, 0)$ so $f(x) = x + 6\pi < 0 \le g(x)$: no solution. Between humps $f < 0$: no solution.

Hence S has $12 + 1 = 13$ elements.

Final Answer: 13

Question 04

English

Let D be the set of all real numbers x for which $\sin^{-1}\left(\frac{[x]}{2}\right) - \cos^{-1}(x - [x])$ is defined, where $[t]$ denotes the greatest integer less than or equal to t and the inverse trigonometric functions take their principal values. Let $f : D \to \mathbb{R}$ be defined by $f(x) = \sin^{-1}\left(\frac{[x]}{2}\right) - \cos^{-1}(x - [x])$. If the range of f is the interval $[\alpha, \beta)$, then the value of $\frac{\beta}{\alpha}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि D उन सभी वास्तविक संख्याओं x का समुच्चय है जिनके लिए $\sin^{-1}\left(\frac{[x]}{2}\right) - \cos^{-1}(x - [x])$ परिभाषित है, जहाँ $[t]$, t से कम या उसके बराबर महत्तम पूर्णांक को दर्शाता है तथा प्रतिलोम त्रिकोणमितीय फलन अपने मुख्य मान लेते हैं। माना कि $f : D \to \mathbb{R}$, $f(x) = \sin^{-1}\left(\frac{[x]}{2}\right) - \cos^{-1}(x - [x])$ द्वारा परिभाषित है। यदि f का परिसर अन्तराल $[\alpha, \beta)$ है, तो $\frac{\beta}{\alpha}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Domain. $x - [x] = \{x\}$, the fractional part, lies in $[0, 1) \subset [-1, 1]$, so $\cos^{-1}(x - [x])$ is defined for every real x. For $\sin^{-1}\frac{[x]}{2}$ we need $-1 \le \frac{[x]}{2} \le 1$, i.e. the integer $[x] \in \{-2, -1, 0, 1, 2\}$, i.e. $-2 \le x < 3$. So $D = [-2, 3)$.

Fix $n = [x] \in \{-2, \ldots, 2\}$. As x runs over $[n, n + 1)$, $t = x - n$ runs over all of $[0, 1)$. Since $\cos^{-1}$ is decreasing on $[0, 1]$ with $\cos^{-1} 0 = \frac{\pi}{2}$ and $\cos^{-1} 1 = 0$, $\cos^{-1} t$ runs over $\left(0, \frac{\pi}{2}\right]$ (the value 0 is excluded because $t = 1$ is excluded).

So on $[n, n + 1)$, $f = s_n - \cos^{-1} t$ takes exactly the values in $\left[s_n - \frac{\pi}{2}, s_n\right)$, where $s_n = \sin^{-1}\frac{n}{2}$: $s_{-2} = -\frac{\pi}{2}$, $s_{-1} = -\frac{\pi}{6}$, $s_0 = 0$, $s_1 = \frac{\pi}{6}$, $s_2 = \frac{\pi}{2}$.

The five pieces are $\left[-\pi, -\frac{\pi}{2}\right)$, $\left[-\frac{2\pi}{3}, -\frac{\pi}{6}\right)$, $\left[-\frac{\pi}{2}, 0\right)$, $\left[-\frac{\pi}{3}, \frac{\pi}{6}\right)$, $\left[0, \frac{\pi}{2}\right)$. Each piece starts before the previous one ends (for example $-\frac{2\pi}{3} < -\frac{\pi}{2}$), so there is no gap and the union is $\left[-\pi, \frac{\pi}{2}\right)$.

The least value $-\pi$ is attained at $x = -2$: $\sin^{-1}(-1) - \cos^{-1} 0 = -\frac{\pi}{2} - \frac{\pi}{2} = -\pi$. The value $\frac{\pi}{2}$ would need $\cos^{-1} t = 0$, i.e. $t = 1$, which is impossible, so it is not attained. Thus $\alpha = -\pi$ and $\beta = \frac{\pi}{2}$.

$\frac{\beta}{\alpha} = \frac{\pi/2}{-\pi} = -\frac{1}{2} = -0.5$.

Final Answer: -0.50

Question 05

English

Let D be the set of all real numbers x for which both $\sin^{-1}\left(\frac{x\left(3-x^2\right)}{2}\right)$ and $\sin^{-1}\left(\frac{x}{2}\right)$ are defined, where $\sin^{-1}$ takes values in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. Let $F : D \to \mathbb{R}$ be defined by $F(x) = \sin^{-1}\left(\frac{x\left(3-x^2\right)}{2}\right) - 3\sin^{-1}\left(\frac{x}{2}\right)$. If m is the minimum value of F on D, then the value of $\frac{m}{\pi}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि D उन सभी वास्तविक संख्याओं x का समुच्चय है जिनके लिए $\sin^{-1}\left(\frac{x\left(3-x^2\right)}{2}\right)$ तथा $\sin^{-1}\left(\frac{x}{2}\right)$ दोनों परिभाषित हैं, जहाँ $\sin^{-1}$ के मान $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ में हैं। माना कि $F : D \to \mathbb{R}$, $F(x) = \sin^{-1}\left(\frac{x\left(3-x^2\right)}{2}\right) - 3\sin^{-1}\left(\frac{x}{2}\right)$ द्वारा परिभाषित है। यदि D पर F का न्यूनतम मान m है, तो $\frac{m}{\pi}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Domain of the first term: we need $-1 \le \frac{x(3-x^2)}{2} \le 1$.

Upper bound: $3x - x^3 \le 2 \iff x^3 - 3x + 2 \ge 0$. Since $x = 1$ is a double root, $x^3 - 3x + 2 = (x-1)^2(x+2)$, which is ≥ 0 exactly when $x \ge -2$.

Lower bound: $3x - x^3 \ge -2 \iff x^3 - 3x - 2 \le 0$, and $x^3 - 3x - 2 = (x+1)^2(x-2)$, which is ≤ 0 exactly when $x \le 2$.

So the first term is defined on $[-2, 2]$; the second needs $|x| \le 2$ as well. Hence $D = [-2, 2]$.

Substitution: for $x \in [-2, 2]$ put $x = 2\sin t$ with $t = \sin^{-1}\frac{x}{2} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. Then $\frac{x(3-x^2)}{2} = \sin t\,(3 - 4\sin^2 t) = \sin 3t$, so $F(x) = \sin^{-1}(\sin 3t) - 3t$ with $3t \in \left[-\frac{3\pi}{2}, \frac{3\pi}{2}\right]$.

Case $-\frac{\pi}{6} \le t \le \frac{\pi}{6}$ ($|x| \le 1$): $3t \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, so $\sin^{-1}(\sin 3t) = 3t$ and $F = 0$.

Case $\frac{\pi}{6} \le t \le \frac{\pi}{2}$ ($1 \le x \le 2$): $\pi - 3t \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ and $\sin(\pi - 3t) = \sin 3t$, so $\sin^{-1}(\sin 3t) = \pi - 3t$ and $F = \pi - 6t$. This decreases from 0 (at $x = 1$) to $\pi - 3\pi = -2\pi$ (at $x = 2$).

Case $-\frac{\pi}{2} \le t \le -\frac{\pi}{6}$ ($-2 \le x \le -1$): $\sin^{-1}(\sin 3t) = -\pi - 3t$ and $F = -\pi - 6t$, which lies between 0 and 2π.

So the minimum is attained at the end point $x = 2$ of the domain. Direct evaluation confirms it: $F(2) = \sin^{-1}(-1) - 3\sin^{-1}(1) = -\frac{\pi}{2} - \frac{3\pi}{2} = -2\pi$.

Hence $m = -2\pi$ and $\frac{m}{\pi} = -2$.

Final Answer: -2

Question 06

English

Let $S = \sum_{k=1}^{\infty} \cot^{-1}\left(\frac{9k^2 - 15k + 5}{3}\right)$, where $\cot^{-1}$ takes values in $(0, \pi)$. Then the value of $\frac{2S}{\pi}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $S = \sum_{k=1}^{\infty} \cot^{-1}\left(\frac{9k^2 - 15k + 5}{3}\right)$ है, जहाँ $\cot^{-1}$ के मान $(0, \pi)$ में हैं। तब $\frac{2S}{\pi}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Look for u_k with $u_{k+1} - u_k$ constant and $1 + u_k u_{k+1}$ equal to the numerator. With $u_k = 3k - 4$: $u_{k+1} = 3k - 1$, $u_{k+1} - u_k = 3$ and $1 + (3k - 4)(3k - 1) = 1 + 9k^2 - 15k + 4 = 9k^2 - 15k + 5$. So the k-th argument is $\frac{1 + u_k u_{k+1}}{u_{k+1} - u_k}$.

Claim: if $b > a$, then $\cot^{-1} \frac{1 + ab}{b - a} = \tan^{-1} b - \tan^{-1} a$. Indeed $\theta = \tan^{-1} b - \tan^{-1} a$ lies in $(0, \pi)$ because $\tan^{-1}$ is increasing with values in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, and $\cot\theta = \frac{1 + \tan\alpha \tan\beta}{\tan\alpha - \tan\beta}$ with $\tan\alpha = b$, $\tan\beta = a$, which is $\frac{1 + ab}{b - a}$. As θ is in the range $(0, \pi)$ of $\cot^{-1}$, the claim follows. No case split is needed, even when $1 + ab < 0$.

For $k = 1$: $u_1 = -1$, $u_2 = 2$, the argument is $\frac{1 - 2}{3} = -\frac{1}{3}$, and the term is $\cot^{-1}\left(-\frac{1}{3}\right) = \tan^{-1} 2 - \tan^{-1}(-1) = \tan^{-1} 2 + \frac{\pi}{4}$, which is greater than $\frac{\pi}{2}$, as it must be for a negative argument. Using $\tan^{-1}(-3)$ here instead would be off by π.

The partial sum telescopes: $\sum_{k=1}^{n} \left[\tan^{-1}(3k - 1) - \tan^{-1}(3k - 4)\right] = \tan^{-1}(3n - 1) - \tan^{-1}(-1) = \tan^{-1}(3n - 1) + \frac{\pi}{4}$.

As $n \to \infty$, $\tan^{-1}(3n - 1) \to \frac{\pi}{2}$, so $S = \frac{3\pi}{4}$.

Therefore $\frac{2S}{\pi} = \frac{3}{2} = 1.5$.

Final Answer: 1.50

Question 07

English

Let $S = \left\{x \in [0, 2\pi] : x \neq \frac{\pi}{2},\ x \neq \frac{3\pi}{2} \text{ and } \cos^{-1}(\cos x) < 3\tan^{-1}(\tan x)\right\}$, where the inverse trigonometric functions take their principal values. If the sum of the lengths of the intervals that constitute S is $k\pi$, then the value of k is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $S = \left\{x \in [0, 2\pi] : x \neq \frac{\pi}{2},\ x \neq \frac{3\pi}{2},\ \cos^{-1}(\cos x) < 3\tan^{-1}(\tan x)\right\}$ है, जहाँ प्रतिलोम त्रिकोणमितीय फलन अपने मुख्य मान लेते हैं। यदि S को बनाने वाले अन्तरालों की लम्बाइयों का योग $k\pi$ है, तो k का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Graph of $C(x) = \cos^{-1}(\cos x)$ on $[0, 2\pi]$: $C(x) = x$ for $0 \le x \le \pi$ and $C(x) = 2\pi - x$ for $\pi \le x \le 2\pi$ (a tent with peak π at $x = \pi$, always ≥ 0).

Graph of $T(x) = \tan^{-1}(\tan x)$: it is $x - n\pi$ where n is chosen so that the value lies in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$; so $T = x$ on $\left[0, \frac{\pi}{2}\right)$, $T = x - \pi$ on $\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$, and $T = x - 2\pi$ on $\left(\frac{3\pi}{2}, 2\pi\right]$.

On $\left[0, \frac{\pi}{2}\right)$: $x < 3x$ holds for $x > 0$, giving $\left(0, \frac{\pi}{2}\right)$ of length $\frac{\pi}{2}$.

On $\left(\frac{\pi}{2}, \pi\right]$: $T = x - \pi \le 0$ while $C = x > 0$, so $C < 3T$ never holds.

On $\left(\pi, \frac{3\pi}{2}\right)$: the inequality $2\pi - x < 3(x - \pi)$ gives $4x > 5\pi$, i.e. $x > \frac{5\pi}{4}$, so the part $\left(\frac{5\pi}{4}, \frac{3\pi}{2}\right)$ of length $\frac{\pi}{4}$.

On $\left(\frac{3\pi}{2}, 2\pi\right]$: $T = x - 2\pi \le 0 \le C$, and at $x = 2\pi$ both are 0, so the strict inequality fails throughout.

Hence $S = \left(0, \frac{\pi}{2}\right) \cup \left(\frac{5\pi}{4}, \frac{3\pi}{2}\right)$, total length $\frac{\pi}{2} + \frac{\pi}{4} = \frac{3\pi}{4}$, and $k = 0.75$.

Final Answer: 0.75

Question 08

English

Let $S = \sum_{k=1}^{7} \cos^{-1}\left(\sin\frac{2k\pi}{9}\right)$, where $\cos^{-1}$ takes values in $[0, \pi]$. Then the value of $\frac{180S}{\pi}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $S = \sum_{k=1}^{7} \cos^{-1}\left(\sin\frac{2k\pi}{9}\right)$ है, जहाँ $\cos^{-1}$ के मान $[0, \pi]$ में हैं। तब $\frac{180S}{\pi}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For $y \in [-1, 1]$, $\sin^{-1} y + \cos^{-1} y = \frac{\pi}{2}$. Hence each term is $\cos^{-1}(\sin\theta_k) = \frac{\pi}{2} - \sin^{-1}(\sin\theta_k)$ with $\theta_k = \frac{2k\pi}{9}$, and $S = \frac{7\pi}{2} - \sum_{k=1}^{7} \sin^{-1}(\sin\theta_k)$.

Work in degrees: $\theta_k = 40k^\circ$, i.e. $40^\circ, 80^\circ, 120^\circ, 160^\circ, 200^\circ, 240^\circ, 280^\circ$. The value $\sin^{-1}(\sin\theta)$ is the unique angle in $[-90^\circ, 90^\circ]$ with the same sine.

40° and 80° are already in range. For $120^\circ, 160^\circ, 200^\circ, 240^\circ$ use $\sin\theta = \sin(180^\circ - \theta)$: $60^\circ, 20^\circ, -20^\circ, -60^\circ$, all in range. For 280° use $\sin\theta = \sin(\theta - 360^\circ)$: -80°.

Sum of these: $40 + 80 + 60 + 20 - 20 - 60 - 80 = 40$ (degrees).

So, in degrees, $S = 7 \cdot 90^\circ - 40^\circ = 590^\circ$, i.e. $\frac{180S}{\pi} = 590$. As a check, the individual terms are $50, 10, 30, 70, 110, 150, 170$ degrees, each in $[0^\circ, 180^\circ]$, with sum 590.

Final Answer: 590

Question 09

English

Considering only the principal values of the inverse trigonometric functions, with $\cot^{-1}$ taking values in $(0, \pi)$, let S be the set of all integers n with $1 \leq |n| \leq 10$ such that $\cot^{-1}(\cot n) = \tan^{-1}(\tan n)$. Then the sum of all the elements of S is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

प्रतिलोम त्रिकोणमितीय फलनों के केवल मुख्य मानों को लेते हुए, जहाँ $\cot^{-1}$ के मान $(0, \pi)$ में हैं, माना कि S उन सभी पूर्णांकों n का समुच्चय है जिनके लिए $1 \leq |n| \leq 10$ तथा $\cot^{-1}(\cot n) = \tan^{-1}(\tan n)$ है। तब S के सभी अवयवों का योग _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let r be the number in $(0, \pi)$ with $n - r$ a multiple of π; since π is irrational, n is never a multiple of π, so $\cot n$ and $\tan n$ are defined and $r \neq \dfrac{\pi}{2}$.

$\cot^{-1}(\cot n)$ is the angle in $(0, \pi)$ with the same cotangent as n; since cot has period π, it equals r. $\tan^{-1}(\tan n)$ is the angle in $\left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right)$ congruent to n modulo π: it is r when $r < \dfrac{\pi}{2}$ and $r - \pi$ when $r > \dfrac{\pi}{2}$. So the two sides are equal exactly when $0 < r < \dfrac{\pi}{2} \approx 1.571$.

Positive n (using $\pi \approx 3.1416$): $r(1) = 1$, $r(2) = 2$, $r(3) = 3$, $r(4) = 4 - \pi \approx 0.858$, $r(5) \approx 1.858$, $r(6) \approx 2.858$, $r(7) = 7 - 2\pi \approx 0.717$, $r(8) \approx 1.717$, $r(9) \approx 2.717$, $r(10) = 10 - 3\pi \approx 0.575$. Those below 1.571: $n = 1, 4, 7, 10$.

Negative n: $r(-n) = \pi - r(n)$. So $r(-1) \approx 2.142$, $r(-2) \approx 1.142$, $r(-3) \approx 0.142$, $r(-4) \approx 2.284$, $r(-5) \approx 1.284$, $r(-6) \approx 0.284$, $r(-7) \approx 2.425$, $r(-8) \approx 1.425$, $r(-9) \approx 0.425$, $r(-10) \approx 2.566$. Those below 1.571: $n = -2, -3, -5, -6, -8, -9$.

So $S = \{1, 4, 7, 10, -2, -3, -5, -6, -8, -9\}$ and the sum is $22 - 33 = -11$.

Final Answer: -11

Question 10

English

Let $f : [-1, 1] \to \mathbb{R}$ be defined by $f(x) = \cos^{-1}\left(\frac{x + \sqrt{3 - 3x^2}}{2}\right) + \cos^{-1}\left(\frac{x - \sqrt{3 - 3x^2}}{2}\right)$, where $\cos^{-1}$ takes values in $[0, \pi]$. Then the value of $\frac{1}{\pi}\sum_{k=0}^{18} f\left(\cos\frac{k\pi}{18}\right)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : [-1, 1] \to \mathbb{R}$, $f(x) = \cos^{-1}\left(\frac{x + \sqrt{3 - 3x^2}}{2}\right) + \cos^{-1}\left(\frac{x - \sqrt{3 - 3x^2}}{2}\right)$ द्वारा परिभाषित है, जहाँ $\cos^{-1}$ के मान $[0, \pi]$ में हैं। तब $\frac{1}{\pi}\sum_{k=0}^{18} f\left(\cos\frac{k\pi}{18}\right)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Substitution: every $x \in [-1, 1]$ is $\cos\theta$ for a unique $\theta \in [0, \pi]$, and then $\sin\theta \geq 0$, so $\sqrt{3 - 3x^2} = \sqrt{3}\sin\theta$. In the sum, $x = \cos\frac{k\pi}{18}$ corresponds to $\theta = \frac{k\pi}{18} \in [0, \pi]$.

Compound angles: $\frac{\cos\theta + \sqrt{3}\sin\theta}{2} = \cos\theta\cos\frac{\pi}{3} + \sin\theta\sin\frac{\pi}{3} = \cos\left(\theta - \frac{\pi}{3}\right)$ and $\frac{\cos\theta - \sqrt{3}\sin\theta}{2} = \cos\left(\theta + \frac{\pi}{3}\right)$.

First term: $\theta - \frac{\pi}{3} \in \left[-\frac{\pi}{3}, \frac{2\pi}{3}\right]$, and $\cos^{-1}(\cos\phi) = |\phi|$ for $\phi \in [-\pi, \pi]$, so it equals $\left|\theta - \frac{\pi}{3}\right|$.

Second term: $\theta + \frac{\pi}{3} \in \left[\frac{\pi}{3}, \frac{4\pi}{3}\right]$. If $\theta \leq \frac{2\pi}{3}$ it is in $[0, \pi]$ and the term is $\theta + \frac{\pi}{3}$; if $\theta > \frac{2\pi}{3}$ it exceeds π and the term is $2\pi - \left(\theta + \frac{\pi}{3}\right) = \frac{5\pi}{3} - \theta$.

Adding the two angles: for $0 \leq \theta \leq \frac{\pi}{3}$, $f = \left(\frac{\pi}{3} - \theta\right) + \left(\theta + \frac{\pi}{3}\right) = \frac{2\pi}{3}$; for $\frac{\pi}{3} \leq \theta \leq \frac{2\pi}{3}$, $f = \left(\theta - \frac{\pi}{3}\right) + \left(\theta + \frac{\pi}{3}\right) = 2\theta$; for $\frac{2\pi}{3} \leq \theta \leq \pi$, $f = \left(\theta - \frac{\pi}{3}\right) + \left(\frac{5\pi}{3} - \theta\right) = \frac{4\pi}{3}$. The pieces agree at $\theta = \frac{\pi}{3}$ and $\theta = \frac{2\pi}{3}$.

Now $\theta = \frac{k\pi}{18}$. For $k = 0, 1, \ldots, 6$ ($\theta \leq \frac{\pi}{3}$) each value is $\frac{2\pi}{3}$: total $\frac{14\pi}{3}$. For $k = 7, \ldots, 11$ the values are $\frac{2k\pi}{18}$: total $\frac{\pi}{9}(7 + 8 + 9 + 10 + 11) = 5\pi$. For $k = 12, \ldots, 18$ ($\theta \geq \frac{2\pi}{3}$) each value is $\frac{4\pi}{3}$: total $\frac{28\pi}{3}$.

Sum $= \frac{14\pi}{3} + 5\pi + \frac{28\pi}{3} = 14\pi + 5\pi = 19\pi$, and dividing by π gives 19.

Final Answer: 19

Question 11

English

Let $u : (-1, 1) \to \mathbb{R}$ and $v : (-1, 1) \to \mathbb{R}$ be defined by $u(x) = \sin^{-1}\left(2x\sqrt{1-x^2}\right)$ and $v(x) = \tan^{-1}\sqrt{\dfrac{1-x}{1+x}}$, where the inverse trigonometric functions take their principal values. There is exactly one $x_0 \in (-1, 1)$ such that $u(x_0) = v(x_0)$. Then the value of the derivative of v with respect to u at $x = x_0$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $u : (-1, 1) \to \mathbb{R}$ तथा $v : (-1, 1) \to \mathbb{R}$, $u(x) = \sin^{-1}\left(2x\sqrt{1-x^2}\right)$ तथा $v(x) = \tan^{-1}\sqrt{\dfrac{1-x}{1+x}}$ द्वारा परिभाषित हैं, जहाँ प्रतिलोम त्रिकोणमितीय फलन अपने मुख्य मान लेते हैं। ठीक एक $x_0 \in (-1, 1)$ ऐसा है कि $u(x_0) = v(x_0)$ है। तब $x = x_0$ पर u के सापेक्ष v के अवकलज का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Substitution: write $x = \cos\phi$ with $\phi \in (0, \pi)$; then $\sin\phi > 0$ and $\sqrt{1-x^2} = \sin\phi$.

For v: $\dfrac{1-\cos\phi}{1+\cos\phi} = \tan^2\dfrac{\phi}{2}$ and $\tan\dfrac{\phi}{2} > 0$, so $v = \tan^{-1}\left(\tan\dfrac{\phi}{2}\right) = \dfrac{\phi}{2}$ because $\dfrac{\phi}{2} \in \left(0, \dfrac{\pi}{2}\right)$.

For u: $2x\sqrt{1-x^2} = 2\cos\phi\sin\phi = \sin 2\phi$ with $2\phi \in (0, 2\pi)$. Bringing 2ϕ into $\left[-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right]$: $u = 2\phi$ if $0 < \phi \le \dfrac{\pi}{4}$; $u = \pi - 2\phi$ if $\dfrac{\pi}{4} \le \phi \le \dfrac{3\pi}{4}$; $u = 2\phi - 2\pi$ if $\dfrac{3\pi}{4} \le \phi < \pi$.

Solving $u = v$. First piece: $2\phi = \dfrac{\phi}{2}$ forces $\phi = 0$, i.e. $x = 1$, which is not in $(-1, 1)$. Second piece: $\pi - 2\phi = \dfrac{\phi}{2}$ gives $\phi = \dfrac{2\pi}{5}$, which lies in $\left[\dfrac{\pi}{4}, \dfrac{3\pi}{4}\right]$. Third piece: $u \le 0 < v$, no solution. So $x_0 = \cos\dfrac{2\pi}{5} = \dfrac{\sqrt{5}-1}{4}$, confirming uniqueness.

Since $\phi_0 = \dfrac{2\pi}{5}$ is strictly inside $\left(\dfrac{\pi}{4}, \dfrac{3\pi}{4}\right)$, near x_0 we have $u = \pi - 2\phi$ and $v = \dfrac{\phi}{2}$, both differentiable in ϕ, and $\phi = \cos^{-1}x$ is differentiable with nonzero derivative there.

Differentiation of one function with respect to another: $\dfrac{dv}{du} = \dfrac{dv/d\phi}{du/d\phi} = \dfrac{1/2}{-2} = -\dfrac{1}{4}$. (Ignoring the branch and taking $u = 2\phi$ gives the wrong value $+\dfrac{1}{4}$.)

Hence the required derivative is -0.25.

Final Answer: -0.25

Question 12

English

Let $x_0 = 0$ and, for every integer $n \geq 0$, let $x_{n+1} = \dfrac{4x_n + 3\sqrt{1 - x_n^2}}{5}$. Then the value of x_{100} is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $x_0 = 0$ है तथा प्रत्येक पूर्णांक $n \geq 0$ के लिए $x_{n+1} = \dfrac{4x_n + 3\sqrt{1 - x_n^2}}{5}$ है। तब x_{100} का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Each x_n lies in $[-1, 1]$ (shown below), so let $\theta_n = \sin^{-1} x_n \in \left[-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right]$; then $\sqrt{1 - x_n^2} = \cos\theta_n \geq 0$.

Let $\varphi = \sin^{-1}\dfrac{3}{5}$, so $\cos\varphi = \dfrac{4}{5}$.

Compound angle: $x_{n+1} = \dfrac{4}{5}\sin\theta_n + \dfrac{3}{5}\cos\theta_n = \sin\theta_n\cos\varphi + \cos\theta_n\sin\varphi = \sin(\theta_n + \varphi)$. In particular $x_{n+1} \in [-1, 1]$.

Addition of inverse sines: $\theta_{n+1} = \sin^{-1}\sin(\theta_n + \varphi)$, which equals $\theta_n + \varphi$ when $\theta_n + \varphi \leq \dfrac{\pi}{2}$ and $\pi - (\theta_n + \varphi)$ when $\dfrac{\pi}{2} < \theta_n + \varphi \leq \dfrac{3\pi}{2}$. (This is the rule $\sin^{-1} a + \sin^{-1} b = \pi - \sin^{-1}\left(a\sqrt{1 - b^2} + b\sqrt{1 - a^2}\right)$ when $a, b > 0$ and $a^2 + b^2 > 1$.)

Since $\tan\varphi = \dfrac{3}{4}$, we have $\varphi < \dfrac{\pi}{4}$, and $\tan 2\varphi = \dfrac{24}{7} > \sqrt{3}$ gives $2\varphi > \dfrac{\pi}{3}$, so $3\varphi > \dfrac{\pi}{2}$. Also $2\varphi < \dfrac{\pi}{2}$.

$\theta_0 = 0$, $\theta_1 = \varphi$, $\theta_2 = 2\varphi$. Then $\theta_2 + \varphi = 3\varphi > \dfrac{\pi}{2}$, so $\theta_3 = \pi - 3\varphi$. Then $\theta_3 + \varphi = \pi - 2\varphi > \dfrac{\pi}{2}$, so $\theta_4 = \pi - (\pi - 2\varphi) = 2\varphi = \theta_2$.

Hence from $n = 2$ the sequence of angles alternates $2\varphi,\ \pi - 3\varphi,\ 2\varphi, \ldots$, and $\theta_n = 2\varphi$ for every even $n \geq 2$.

Therefore $x_{100} = \sin 2\varphi = 2\sin\varphi\cos\varphi = 2 \cdot \dfrac{3}{5} \cdot \dfrac{4}{5} = \dfrac{24}{25} = 0.96$. (Check: $x_1 = 0.6$, $x_2 = 0.96$, $x_3 = \dfrac{117}{125} = 0.936$, $x_4 = 0.96$.)

Final Answer: 0.96

Question 13

English

Considering only the principal value of $\tan^{-1}$, the number of ordered pairs (m, n) of positive integers with $m \leq 50$ and $n \leq 50$ satisfying $\tan^{-1}\frac{1}{m} + \tan^{-1}\frac{1}{n} \geq \tan^{-1}\frac{1}{2}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$\tan^{-1}$ के केवल मुख्य मान को लेते हुए, धनात्मक पूर्णांकों के उन क्रमित युग्मों (m, n) की संख्या, जिनके लिए $m \leq 50$, $n \leq 50$ तथा $\tan^{-1}\frac{1}{m} + \tan^{-1}\frac{1}{n} \geq \tan^{-1}\frac{1}{2}$ है, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Case 1: $m \leq 2$ or $n \leq 2$. If $m \in \{1, 2\}$ then $\tan^{-1}\frac{1}{m} \geq \tan^{-1}\frac{1}{2}$ (as $\tan^{-1}$ is increasing), and $\tan^{-1}\frac{1}{n} > 0$, so the inequality holds strictly. The same holds if $n \in \{1, 2\}$. The number of pairs in $\{1, \ldots, 50\}^2$ with $m \leq 2$ or $n \leq 2$ is $50^2 - 48^2 = 2500 - 2304 = 196$.

Case 2: $m, n \geq 3$. Then $\frac{1}{m} \cdot \frac{1}{n} < 1$, so the addition formula applies without correction: $\tan^{-1}\frac{1}{m} + \tan^{-1}\frac{1}{n} = \tan^{-1}\frac{\frac{1}{m} + \frac{1}{n}}{1 - \frac{1}{mn}} = \tan^{-1}\frac{m+n}{mn-1}$, an angle in $\left(0, \frac{\pi}{2}\right)$.

Because $\tan^{-1}$ is increasing, the inequality becomes $\frac{m+n}{mn-1} \geq \frac{1}{2}$, i.e. $2m + 2n \geq mn - 1$ (as $mn - 1 > 0$), i.e. $mn - 2m - 2n + 4 \leq 5$, i.e. $(m-2)(n-2) \leq 5$.

The boundary is the equation $\tan^{-1}\frac{1}{m} + \tan^{-1}\frac{1}{n} = \tan^{-1}\frac{1}{2}$, i.e. $(m-2)(n-2) = 5$, whose solutions with $m, n \geq 3$ are $(3, 7)$ and $(7, 3)$; these satisfy the non-strict inequality and are counted.

With $a = m-2 \geq 1$ and $b = n-2 \geq 1$, the number of pairs with $ab \leq 5$ is $\sum_{a=1}^{5} \left[\frac{5}{a}\right] = 5+2+1+1+1 = 10$: they are $(1, b)$ for $b \leq 5$, $(2, 1), (2, 2)$, $(3, 1)$, $(4, 1)$, $(5, 1)$. All have $m, n \leq 7 \leq 50$.

Total: $196 + 10 = 206$.

Final Answer: 206

Question 14

English

Let S be the set of all real numbers x satisfying $\sin\left(2\tan^{-1}x\right) = \cos\left(2\tan^{-1}\dfrac{2x}{3}\right)$, where $\tan^{-1}$ takes values in $\left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right)$. Then the sum of the squares of all the elements of S is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि S उन सभी वास्तविक संख्याओं x का समुच्चय है जो $\sin\left(2\tan^{-1}x\right) = \cos\left(2\tan^{-1}\dfrac{2x}{3}\right)$ को संतुष्ट करती हैं, जहाँ $\tan^{-1}$ के मान $\left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right)$ में हैं। तब S के सभी अवयवों के वर्गों का योग _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $\alpha = \tan^{-1}x$ and $\beta = \tan^{-1}\dfrac{2x}{3}$. Both lie in $\left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right)$ and have the sign of x, so $|\alpha - \beta| < \dfrac{\pi}{2}$ and $\alpha + \beta \in (-\pi, \pi)$.

Trigonometric equation: $\sin 2\alpha = \cos 2\beta$ means $\cos\left(\dfrac{\pi}{2} - 2\alpha\right) = \cos 2\beta$, whose general solution is $\dfrac{\pi}{2} - 2\alpha = 2k\pi \pm 2\beta$, $k \in \mathbb{Z}$, i.e. $\alpha + \beta = \dfrac{\pi}{4} - k\pi$ or $\alpha - \beta = \dfrac{\pi}{4} - k\pi$.

Family $\alpha + \beta = \dfrac{\pi}{4} - k\pi$ in $(-\pi, \pi)$: only $\dfrac{\pi}{4}$ $(k = 0)$ and $-\dfrac{3\pi}{4}$ $(k = 1)$. In both cases $\tan(\alpha + \beta) = 1$, i.e. $\dfrac{x + \frac{2x}{3}}{1 - \frac{2x^2}{3}} = 1$ (with $2x^2 \neq 3$), so $5x = 3 - 2x^2$, $2x^2 + 5x - 3 = 0$, $(2x - 1)(x + 3) = 0$: $x = \dfrac{1}{2}$ or $x = -3$.

Which angle each gives: for $x = \dfrac{1}{2}$, $\alpha, \beta > 0$ and $\tan^{-1}\dfrac{1}{2} + \tan^{-1}\dfrac{1}{3} = \dfrac{\pi}{4}$; for $x = -3$, $\alpha + \beta = -\left(\tan^{-1}3 + \tan^{-1}2\right) = -\left(\pi - \dfrac{\pi}{4}\right) = -\dfrac{3\pi}{4}$. Both are members of the family, so both are solutions. (Rejecting $x = -3$ because $\alpha + \beta \neq \dfrac{\pi}{4}$ is the trap; the general solution includes $-\dfrac{3\pi}{4}$.)

Family $\alpha - \beta = \dfrac{\pi}{4} - k\pi$: as $|\alpha - \beta| < \dfrac{\pi}{2}$, only $\alpha - \beta = \dfrac{\pi}{4}$, so $\dfrac{x - \frac{2x}{3}}{1 + \frac{2x^2}{3}} = 1$, i.e. $x = 3 + 2x^2$, $2x^2 - x + 3 = 0$, with discriminant $1 - 24 < 0$: no real solution.

Direct check: $x = \dfrac{1}{2}$: $\sin(2\alpha) = \dfrac{2x}{1 + x^2} = \dfrac{4}{5}$ and $\cos 2\beta = \dfrac{1 - 4x^2/9}{1 + 4x^2/9} = \dfrac{8/9}{10/9} = \dfrac{4}{5}$. $x = -3$: $\dfrac{-6}{10} = -\dfrac{3}{5}$ and $\dfrac{1 - 4}{1 + 4} = -\dfrac{3}{5}$.

So $S = \left\{\dfrac{1}{2}, -3\right\}$ and the sum of squares is $\dfrac{1}{4} + 9 = 9.25$.

Final Answer: 9.25